

Digital Logics

BCA — Semester 1 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Define BCD code. Design BCD to Excess-3 Code converter with truth table and logic diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define combinational logic circuit. Design a combinational circuit whose input is a four-bit number and whose output is the 1's complement of the input number.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define parallel counter. Design MOD-12 synchronous up counter along with state diagram, timing sequence and timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Convert (591.62)₁₀ into hexadecimal and octal number system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Why NAND and NOR gates are called Universal logic gates? Realize NOR as Universal logic gates.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Simplify: $F = XYZ' + XZ' + X'Y + X''YZ'$ using K-Map in both SOP and POS.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define PLDs. Implement the given functions with ROM. $F_1 = \sum(0,1,2)$ $F_2 = \sum(0,2,3)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Describe the clocked master-slave J-K flip-flop with its operation table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is forbidden state in SR flip flop? Convert SR to JK flip flop.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Differentiate between ring counter and Johnson counter.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain the operation of 4-bit Parallel-In parallel-Out Shift register with data input 1010.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: (Any two) a. ASCII Code b. Logic micro operation c. State table

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.